Conditioned States as Toric Kähler Data

A companion note to "Probability, Context and Differentiation: The Layers"
(working draft)

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Abstract

The central point of this note is that *leading-order balance identifies candidates, not the selected state*. Let Q be a compact Delzant polytope with potential $W_T = \sum_k T^{-\lambda_k} y^{v_k}$. The unique positive critical point y_T (Galkin's lemma) has a valuation u^*, and at u^* the leading terms of W_T balance with positive weights: Birch's theorem at target 0. The converse fails. On the Hirzebruch surface F_1 the leading terms balance along a whole segment, and higher-order terms select one point of it. Critical points that are not positive give only a signed or complex linear dependence, not a law. The positive orthant is the ascending Lefschetz thimble of y_T, so $e^{-W/\hbar}\, dy/y$ on it is a probability law. We prove a tropical limit for this law, with $\hbar = T^h$ and in the coordinates $\log y / \log T$. Below the max–min level h^* it tends to the uniform law on the shrunk polytope Q_h, which is the Duistermaat–Heckman measure when $h = 0$. At and above h^* it concentrates on $\operatorname{argmax} \min_k \ell_k$, a set that contains every candidate and u^*. On F_1 the mass at level h^* is uniform on the candidate segment, and the selection appears only at the next max–min level. Around these results the note records the classical toric Kähler dictionary in probabilistic terms: the Fisher metric, relative entropy, Duistermaat–Heckman, and the positive Lagrangian section. For Fano X_Q, the Floer theorem of Cho and Oh, cited and not proved, turns this into a statement about torus fibres. Everything is worked exactly on $\mathbb{CP}^2$ and F_1. Finally, soft evidence of size n shifts the exponents, $\lambda \mapsto \lambda - (n/N)\rho$. The max–min level is a linear programme over the laws on the facets with mean exponent zero, and data can change the leading structure only along the circuits of $\{v_k\}$. $\mathbb{CP}^2$ has none. F_1 has one, with its wall at $\alpha = 1/3$, inside the Kähler cone. There the selected valuation moves continuously, while the leading set, the dimension of the candidate set and the mechanism of selection all jump. Part II reads this back on contingency tables. A log-linear model carries two configurations, its cells, whose relations are the moves between tables, and its facets, which build W. The $r \times c$ independence model has $(r-1)(c-1)$ moves but exactly one circuit of facets. Its wall is at $cL_1 = rL_2$, which square tables sit on and other tables do not. Its selected fibre never moves, so its wall, unlike that of F_1, changes no Floer data. Finally, one 2×3 coupling polytope is followed through every layer, with pictures. Its Tier A law is Fisher's noncentral hypergeometric family. Its selected coupling is the leximin one, because the best smallest cell is $\min(f_i/m,\, g_j/n)$. Its positive thimble interpolates from the uniform law on couplings to that coupling. And moving one count across a leximin wall, inside a Kähler chamber, changes its leading cells and its Floer fibres. A ledger records every operation with its status and its checks; 30 numerical checks and 3 ledger checks corroborate the statements.

Part I
The geometry

1 What this note does and does not do

The layers paper ends its toric section with an outlook remark: the precise dictionary is between conditioned states and moment fibres, not between sources and Lagrangians [6, Remark "outlook: a symplectic reading"]. This note makes that remark exact in two places and leaves the rest open. Its components have different status.

component	status	source
toric Kähler structure, moment map, Fisher metric, symplectic potential, Duistermaat–Heckman	Used	[23, 1, 13, 2]
Birch's theorem; relative interiors of convex hulls	Used	[7, 28]
one positive critical point (theorem 3(1)); the positive locus as a Lefschetz thimble	Used	Galkin's lemma [19]; [20, 25]
the valuation u^* maximises $\min_k \ell_k$; a recursive construction of u^*	Used	[26, Thm. 1.2]; [17]
the bridge (theorem 3(2)–(4)): leading-order balance, candidates versus selection, the sign split	Derived	this note
the orthant is the ascending thimble J_+; Laplace concentration (proposition 7)	Adapted	[19, 20]
the tropical limit of the thimble law (theorem 8) and its F_1 levels (example 9)	New	this note
linear-programming duality, strict complementarity; regular subdivisions and secondary fans	Used	[29, 22, 27]
soft evidence as $\lambda - s\rho$; the balance programme; candidates = relative interior of argmax (definition 10 and proposition 11)	Derived	this note
where data can act: circuits, least evidence, the F_1 wall (corollary 12 and example 14)	New	this note
coupling polytopes are Delzant; Kähler class = product coupling; chambers and blow-ups	Used	[4]
Tier A on the fibre is Fisher's noncentral hypergeometric family (proposition 22)	Adapted	[16, 8]
the max–min coupling $\min(f_i/m, g_j/n)$; leximin walls; the one-count example (theorem 23 and example 25)	New	this note
Floer cohomology of toric fibres and the potential W	Cited, not proved	[10, 17]
Novikov weights against the environment orders of the layers paper; decision, contextuality and VOI walls	Open	section 12

Two probability structures must be kept apart. The measures in this note live on *conditioned states*, points of the polytope. They are not the source weights π_l of the layers paper. The uniform measure of proposition 1(5) is a prior on states, not on sources.

2 Tier A: the toric Kähler structure of conditioned states

Setting. Let $S = \{S_i\} \subset \mathbb{Z}^d$ be a finite support whose affine span is $\mathbb{R}^d$, with coefficients $c_i > 0$. In log coordinates $u = \log x$ put

$$\varphi(u) = \log \sum_i c_i e^{\langle S_i, u\rangle}, \qquad q_i(u) = \frac{c_i e^{\langle S_i, u\rangle}}{\sum_j c_j e^{\langle S_j, u\rangle}},$$

so $q(u)$ is the monomial law of the layers paper and $\mu(u) = \sum_i q_i S_i$ is its moment map. On the torus $(\mathbb{C}^*)^d$, write $z_j = e^{(u_j + i\theta_j)/2}$.

Proposition 1 (the toric Kähler data)**.** *(1) $F(z) = \varphi(\log|z|^2)$ is a Kähler potential for a torus-invariant Kähler form ω on $(\mathbb{C}^*)^d$, the pullback of the Fubini–Study form by the monomial map $z \mapsto [\sqrt{c_i}\, z^{S_i}]$. In the coordinates (u, θ), up to the normalising constant of one's convention, $\omega = \sum_{j,k} \varphi_{jk}(u)\, du_j \wedge d\theta_k$, and the moment map of the rotation action is $\nabla\varphi = \mu$.*

(2) Fisher information is the metric. *The Hessian $\varphi_{jk} = \mathrm{Cov}_{q(u)}(S)_{jk}$ is the Fisher information of the exponential family $q(u)$ in its natural parameter, and the Kähler metric is $\sum \varphi_{jk}(du_j du_k + d\theta_j d\theta_k)$.*

(3) The positive real locus is a Lagrangian section. *$\{\theta = 0\}$ is Lagrangian, and it meets each moment fibre $\mu^{-1}(p)$, p in the interior of $Q = \mathrm{conv}(S)$, in exactly one point (Birch).*

(4) The symplectic potential is a relative entropy. *The Legendre dual $\psi(p) = \sup_u(\langle p, u\rangle - \varphi(u))$ equals $\sum_i q_i \log(q_i/c_i)$ evaluated at the exponential-family law $q = q(u(p))$. That law is the unique minimiser of $D(q\|c) = \sum_i q_i \log(q_i/c_i)$ among the laws on S with mean p; many other laws have mean p. So $\psi(p) = \min\{D(q\|c) : \sum_i q_i S_i = p\}$, the entropy of the minimiser relative to the weights c, with the sign reversed. Moreover $\nabla\psi(p) = u(p)$ and $\mathrm{Hess}\,\psi = (\mathrm{Hess}\,\varphi)^{-1}$, the Fisher information in the expectation parameter.*

(5) Duistermaat–Heckman. *The pushforward of the Liouville measure $\omega^d/d!$ by μ is $(2\pi)^d$ times Lebesgue measure on the interior of Q. In natural parameters it has density* $\det \mathrm{Hess}\,\varphi$.

Proof. These are classical. For (1), see [23, 1] and [18, Section 4.2]. (2) holds because φ is the log-partition function of the exponential family [2]. (3): on $\{\theta = 0\}$ every $d\theta_k$ vanishes, and a moment fibre is $\{u(p)\} \times T^d$, which meets $\theta = 0$ once because $u(p)$ is unique [7]. (4) is Legendre duality for a strictly convex function, and $\langle p, u\rangle - \varphi(u) = \sum_i q_i \log(q_i/c_i)$ at $\mu(u) = p$. That $q(u(p))$ minimises $D(\cdot\|c)$ under the mean constraint is the Lagrange (Gibbs) characterisation of exponential families [2]. (5) is the change of variables $p = \nabla\varphi(u)$, whose Jacobian is $\mathrm{Hess}\,\varphi$ [13]; the θ-integral gives $(2\pi)^d$. Checks: KB1 checks (1)–(2) on 30 random supports, against finite differences. KB2 checks (4) on 20 supports. KB3 checks (5) by comparing $\int \det \mathrm{Hess}\,\varphi\, du$ over boxes with the areas of their moment images. □

In general this structure lives on the open orbit $(\mathbb{C}^*)^d$ only. A Delzant polytope Q determines a compact symplectic toric manifold X_Q [11], but a metric built from an arbitrary support need not extend smoothly to it. It does extend in one case. Suppose S is the set of lattice points of a Delzant lattice polytope Q. On a smooth projective toric variety an ample line bundle is very ample [18, Section 3.4]. So the monomial map extends to an embedding of X_Q in $\mathbb{P}^{|S|-1}$, and ω is the restriction of the Fubini–Study form; the weights c_i act by a diagonal automorphism. The polytope of the layers paper's blueprint is not Delzant: its vertex at $(1, 1)$ has determinant 3. So the blueprint gives an orbifold, and the examples below are $\mathbb{CP}^2$ and F_1.

3 The bridge

Setting. Let $Q = \{p : \ell_k(p) = \langle v_k, p\rangle - \lambda_k \geq 0,\ k = 1, \dots, m\}$ be a compact Delzant polytope, with primitive inward facet normals v_k. Its potential, in the convention of Hori–Vafa and Cho–Oh [24, 10], is

$$W_T(y) = \sum_k T^{-\lambda_k} y^{v_k}, \qquad y \in (\mathbb{C}^*)^n,\ 0 < T < 1.$$

A family of critical points y_T has *valuation* $u = \lim_{T\to 0} \log|y_T| / \log T$ when the limit exists. At u, the term k has order $\ell_k(u)$, and the *leading terms* are $K(u) = \{k : \ell_k(u) \text{ minimal}\}$. Say the leading terms *balance* at u if $0 = \sum_{k\in K(u)} a_k v_k$ for some $a_k > 0$.

Lemma 2 (balance is Birch at target zero)**.** *For vectors v_k, $k \in K$, the following are equivalent:*

- *there are weights $a_k > 0$ with $\sum_k a_k v_k = 0$;*
- *there is a law q with $q_k > 0$ for every $k \in K$ and mean $\sum_k q_k v_k = 0$;*
- *0 lies in the relative interior of $\operatorname{conv}\{v_k : k \in K\}$.*

When they hold, the positive monomial law with support $\{v_k\}$ and mean 0 is unique, up to the torus directions orthogonal to the span of the differences (Birch).

Proof. Normalising the weights gives the first equivalence. The relative interior of the convex hull of finitely many points is the set of their convex combinations with all coefficients positive [28, Theorem 6.9]. Uniqueness is Birch's theorem [7], which is [6, Theorem "the toric realization"(1)] with target 0. □

Theorem 3 (the Birch bridge)**.** *Let the λ_k be rational.*

(1) One positive critical point *(Galkin's lemma [19]; Used). For each $T \in (0,1)$, W_T has exactly one critical point y_T in the positive torus $(0,\infty)^n$. A positive y is critical iff the law $w_k \propto T^{-\lambda_k} y^{v_k}$ of the terms of W_T has mean exponent $\sum_k w_k v_k = 0$: Birch at target 0.*

(2) Its valuation balances. *The valuation u^* of y_T exists. As $T \to 0$ the law w tends to a law supported exactly on $K(u^*)$, with positive weights and mean 0. So the leading terms balance at u^*.*

(3) Candidates versus selection. *The converse of (2) fails: the set of points where the leading terms balance can be larger than $\{u^*\}$, and higher-order terms select u^* among them.*

(4) The sign split. *At the valuation of a critical point that is not positive, the leading coefficients ρ_k of the surviving terms give a linear dependence $0 = \sum_K \rho_k v_k$ with every $\rho_k \neq 0$, real of either sign or complex. Unlike the positive case, this dependence does not define a probability law, and the leading terms there need not balance.*

Proof. (1) In log coordinates $b = \log y$, $\log W_T(e^b) = \log \sum_k T^{-\lambda_k} e^{\langle v_k, b\rangle}$ is strictly convex. Its Hessian is the covariance of the v_k under w, which is positive definite because the v_k *affinely* span $\mathbb{R}^n$. Indeed the v_k of a compact polytope positively span $\mathbb{R}^n$, so 0 lies in the interior of $\operatorname{conv}\{v_k\}$, which is therefore full-dimensional. Positive spanning also makes the function coercive: every direction increases some term. It therefore has exactly one critical point. Since $W_T > 0$ there, a critical point of $\log W_T$ is a critical point of W_T, and its gradient is the mean of w. (2) Write $T = \tau^N$, with N a common denominator of the λ_k. The positive critical point is a semialgebraic

function of τ, so it has a Puiseux expansion with positive leading coefficients [9], and its valuation exists. The weights w_k then have orders $\ell_k(u^*)$ in T, up to a common shift, so they concentrate on $K(u^*)$. The limits are strictly positive on every $k \in K(u^*)$, namely $w_k \to \rho_k / \sum_{j \in K(u^*)} \rho_j$, where the $\rho_k > 0$ are the leading coefficients of the positive Puiseux series $T^{-\lambda_k} y_T^{v_k}$. That strict positivity is what places 0 in the relative interior. This is the arc-limit theorem of the layers paper. The mean stays 0 in the limit; apply lemma 2. Judd and Rietsch prove that the valuation satisfies tropical critical conditions of this kind for complete positive Laurent polynomials over generalised Puiseux series [26]. (3) is example 6(b). (4) The same expansion, with complex leading coefficients ρ_k, gives $0 = \sum_K \rho_k v_k$ with every $\rho_k \neq 0$. Example 6(a) shows a real critical point with one negative coordinate, whose leading terms admit no positive balance. The checks below corroborate these statements; they are not proofs. KB4 checks (1)–(2) on $\mathbb{CP}^2$ and on F_1 at five blow-up sizes, from four starts each. KB5 checks (3) at exact rational points. KB6 finds all four critical points of F_1 in 80-digit arithmetic and checks (4). □

4 The Floer reading

For a compact toric Fano manifold X_Q, Cho and Oh show that the potential of the Lagrangian torus fibres is exactly W, with no corrections, and that a fibre $L(u)$ has non-vanishing Floer cohomology for some local system iff a critical point of W has valuation u [10]. Fukaya, Oh, Ohta and Ono extend the picture to all compact toric manifolds, where W acquires corrections [17]. We use only the Fano statement, and we prove nothing about Floer cohomology. Both examples below are Fano: $\mathbb{CP}^2$, and F_1, the blow-up of $\mathbb{CP}^2$ at a point, which is a del Pezzo surface. The Hirzebruch surfaces F_n with $n \geq 2$ are not Fano.

Corollary 4 (the probabilistic fibre)**.** *On a compact toric Fano manifold, exactly one critical point of W is positive. Its valuation determines* one *torus fibre $L(u^*)$ with non-vanishing Floer cohomology, not the only one. At u^*, the terms of W of least order form a conditioned law with mean exponent* 0*. Other critical points, including non-positive ones, may determine further fibres with non-vanishing Floer cohomology, but those are not represented by the positive probabilistic law. The fibre $L(u^*)$, a Lagrangian torus, is a different object from the positive real Lagrangian section of proposition 1(3), which meets it once.*

Proof. Combine theorem 3(1)–(2) with the cited theorem. □

Example 5 ($\mathbb{CP}^2$)**.** Here $v = (1,0), (0,1), (-1,-1)$ and $\lambda = (0,0,-1)$, so $W_T = y_1 + y_2 + T/(y_1 y_2)$. The critical points are $y_1 = y_2$ with $y_1^3 = T$: three of them, one positive, all with valuation $(\frac{1}{3}, \frac{1}{3})$, the barycentre. There all three terms tie, and $\frac{1}{3}\big((1,0) + (0,1) + (-1,-1)\big) = 0$. The barycentre is also the only point where the leading terms balance. Its fibre is the Clifford torus.

Example 6 (the Hirzebruch surface F_1)**.** Take $Q = \{p_1 \geq 0,\ p_2 \geq 0,\ p_1 + p_2 \leq 1,\ p_2 \leq 1 - \alpha\}$, the triangle of $\mathbb{CP}^2$ with the corner $(0,1)$ cut by a facet of length $\alpha \in (0,1)$. So $v = (1,0), (0,1), (-1,-1), (0,-1)$ and $W_T = y_1 + y_2 + T/(y_1 y_2) + T^{1-\alpha}/y_2$. The critical equations are $y_1^2 y_2 = T$ and $y_2^2 = T/y_1 + T^{1-\alpha}$. With $y_2 = w^2$ and $y_1 = T^{1/2}/w$, they become $w^4 - T^{1/2} w - T^{1-\alpha} = 0$, so there are four critical points. The Newton polygon of this quartic gives their valuations.

(a) $\alpha < \frac{1}{3}$. Three critical points lie over $(\frac{1}{3}, \frac{1}{3})$, one of them positive, and there the first three terms balance. The fourth lies over $(\alpha, 1 - 2\alpha)$. Its leading coefficient is $w \approx -T^{1/2-\alpha}$, so $y_1 < 0$. Its leading terms are $(1,0), (-1,-1), (0,-1)$. They admit the signed balance

$(1,0)+(-1,-1)-(0,-1)=0$ (in general $a_1=a_3=-a_4$), but no positive one. By corollary 4, the probabilistic fibre is over the barycentre, and the second fibre that the cited theorem provides is outside the positive theory.

(b) $\alpha>\frac{1}{3}$. All four critical points lie over $u^*=(\frac{1+\alpha}{4},\frac{1-\alpha}{2})$, one of them positive. But the leading terms $(0,1)$ and $(0,-1)$ balance at every point of the open segment $u_2=\frac{1-\alpha}{2}$, $\frac{1-\alpha}{2}<u_1<\alpha$. The endpoints are excluded. There a third term joins the leading set, which is $\{v_1,v_2,v_4\}$ at $u_1=\frac{1-\alpha}{2}$ and $\{v_2,v_3,v_4\}$ at $u_1=\alpha$. Neither set balances positively, because the first coordinates of its vectors are $1,0,0$, or $0,-1,0$. Leading order gives a segment of candidates, and the next order selects u^*: the same mechanism by which the layers paper's ties at the limit are broken at higher order.

At $\alpha=\frac{1}{3}$ all four facets tie at the barycentre. KB5 and KB6 corroborate every statement here at $\alpha=\frac{1}{5}$ and $\frac{3}{5}$; KB4 also uses $\frac{1}{4},\frac{2}{5},\frac{4}{5}$ (fig. 1).

5 The positive thimble and its tropical limit

The positive critical point has a second, analytic role: it is the critical point of an oscillatory integral. In mirror symmetry the integral $\int e^{-W/\hbar}\,dy/y$ over the positive real cycle is the one tied to Galkin's conifold point and to Gamma conjecture I; under homological mirror symmetry that cycle is expected to correspond to the structure sheaf [19, 20, 25]. Here we only need its elementary properties, and what the tropical limit does to the law it carries. Give $(\mathbb{C}^*)^n$ the flat metric $\sum_j|d\log y_j|^2$, and let $\hbar>0$.

Proposition 7 (the positive thimble)**.** *(1) The positive orthant $(0,\infty)^n$ is the ascending Lefschetz thimble J_+ of y_T for $\operatorname{Re}(W_T/\hbar)$, the set of points whose downward gradient flow tends to y_T. In $b=\log y$, the Hessian of $\operatorname{Re}W_T(e^b)$ at $b_T=\log y_T$, in the coordinates $(\operatorname{Re}b,\operatorname{Im}b)$, is $\operatorname{diag}(H,-H)$ with $H=\operatorname{Hess}(W_T\circ\exp)(b_T)$ positive definite. So the real directions ascend, and $\operatorname{Im}W_T=0$ on J_+.*

(2) $\mu_\hbar=Z^{-1}e^{-W_T(y)/\hbar}\,dy/y$ on J_+ is a probability law, and $Z<\infty$.

(3) Laplace. *As $\hbar/W_T(y_T)\to 0$, the law of $b=\log y$ under $\mu_\hbar$ is asymptotically normal, with mean b_T and covariance $\hbar H^{-1}$.*

Proof. (1) $W_T\circ\exp(b)=\sum_k T^{-\lambda_k}e^{\langle v_k,b\rangle}$ is holomorphic, and its restriction to real b is strictly convex and coercive (theorem 3(1)). For a holomorphic f, the Hessian of $\operatorname{Re}f$ in $(\operatorname{Re}b,\operatorname{Im}b)$ is $\operatorname{diag}(H,-H)$ with H the real Hessian, by the Cauchy–Riemann equations. The metric is real, so the gradient flow of $\operatorname{Re}W_T$ preserves the real locus. There the flow is the gradient flow of a strictly convex, coercive function, and every orbit tends to b_T. So the orthant lies in the stable manifold J_+, which is an n-ball. The orthant is an n-manifold, so it is open in J_+; it is closed in $(\mathbb{C}^*)^n$, so it is closed in J_+. Since J_+ is connected, the two are equal. (2) $W_T>0$ on J_+, and $W_T(e^b)\geq c|b|-C$ by positive spanning, so $Z<\infty$. (3) This is Laplace's method at the non-degenerate minimum b_T of $W_T\circ\exp$; the relative corrections are $O(\hbar/W_T(y_T))$. KB7 corroborates (1) and (3), at $\hbar/W_T(y_T)=10^{-2},10^{-3},10^{-4}$. □

Laplace describes the limit $\hbar\to 0$ at fixed T. The tropical limit couples the two, $\hbar=T^h$, and reads y on the scale of the valuation, $u=\log y/\log T$. Since $T^{-\lambda_k}y^{v_k}=T^{\ell_k(u)}$, the pushforward

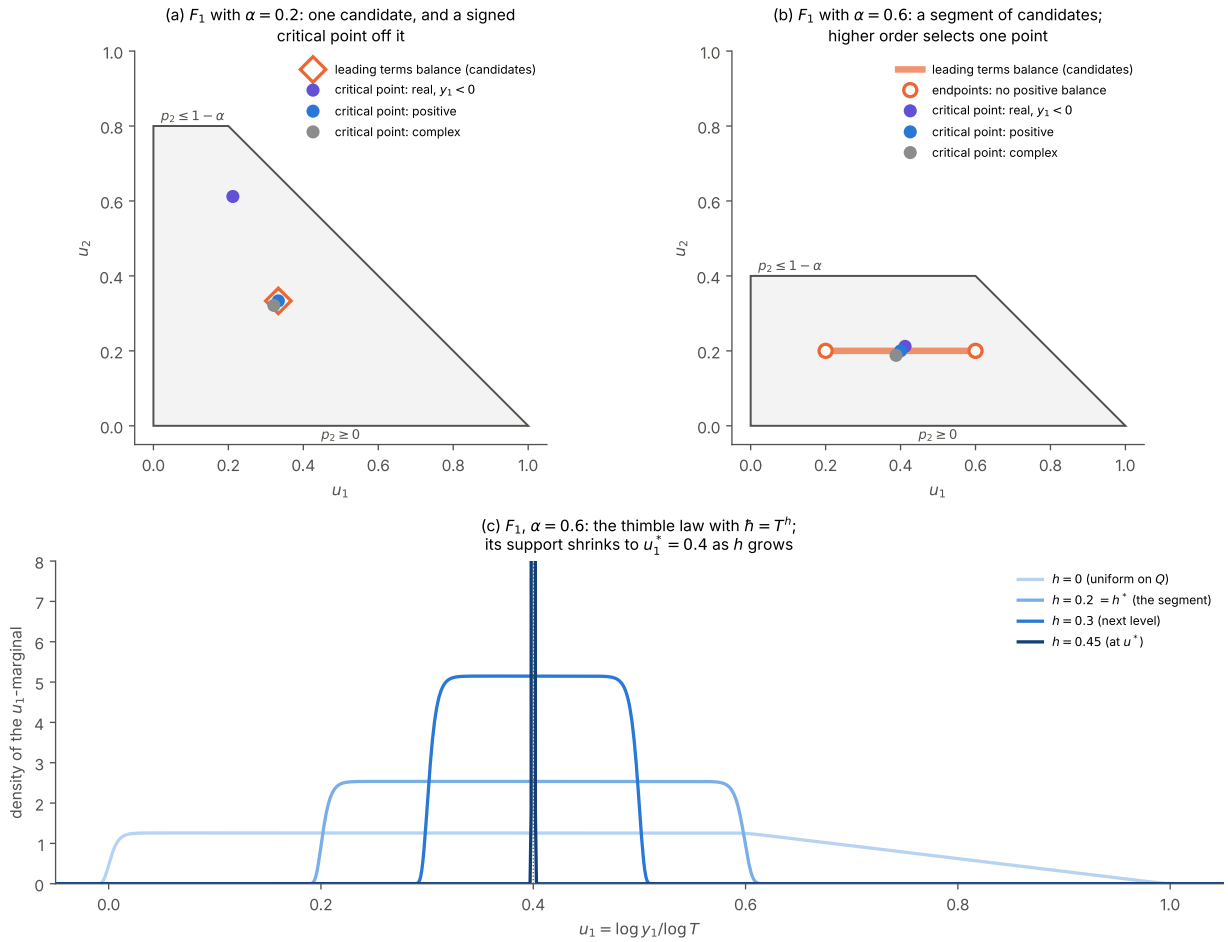


Figure 1: F_1 at two blow-up sizes. (a) $\alpha = 0.2$: the only point where the leading terms balance is the barycentre, which carries the positive critical point; a real critical point with $y_1 < 0$ sits over $(\alpha, 1 - 2\alpha)$, outside the positive theory. (b) $\alpha = 0.6$: the leading terms balance along an open segment, whose endpoints (open circles) have no positive balance, and the critical points all sit over one point of it. (c) $\alpha = 0.6$: the u_1-marginal of the thimble law with $\hbar = T^h$ (theorem 8 and example 9). At $h = 0$ it is the marginal of the uniform law on Q; at $h = h^* = 0.2$ it is uniform on the segment; at $h = 0.3$ it is uniform on $[0.3, 0.5]$; above 0.4 it is concentrated at $u_1^* = 0.4$. Here $\log T = -200$, and -100 for $h = 0.45$.

$\nu_{T,h}$ of μ_{T^h} to u has density proportional to

$$f_{T,h}(u) = \exp\Big(-\sum_k T^{\ell_k(u)-h}\Big).$$

Let $g(u) = \min_k \ell_k(u)$, which is the tropicalisation of W in the min convention. Let $h^* = \max_u g$, and let $Q_h = \{u : g(u) \geq h\} = \{\langle v_k, u\rangle \geq \lambda_k + h\}$, so that $Q_0 = Q$.

Theorem 8 (the tropical limit of the thimble law)**.** *(1) If $h < h^*$, then $\nu_{T,h}$ tends, in total variation, to the uniform law on the polytope Q_h. At $h = 0$ this is the normalised Duistermaat–Heckman measure of X_Q.*

(2) If $h \geq h^*$, then $\nu_{T,h}$ concentrates on $Q_{h^*} = \operatorname{argmax} g$: $\nu_{T,h}(U) \to 1$ for every open $U \supseteq Q_{h^*}$.

(3) $\operatorname{argmax} g = \{u : 0 \in \operatorname{conv}\{v_k : k \in K(u)\}\}$. *It is a polytope. It contains every candidate, every u with $0 \in \operatorname{relint}\operatorname{conv}\{v_k : k \in K(u)\}$, and in particular u^*; it can also contain non-candidates. Every u outside it has $g(u) < h^*$. This applies to the valuation of a signed critical point of example 6(a) as well.*

Proof. Write $L = \log(1/T)$. Since the v_k positively span, $g(u) \leq C - c|u|$, so h^* is finite and attained, and the level sets of g are compact. (1) For u with $g(u) > h$, every exponent is positive and $f_{T,h}(u) \to 1$. Off Q_h, some exponent is negative and $f_{T,h}(u) \to 0$, and ∂Q_h is null. For $T \leq T_0 < 1$, $f_{T,h} \leq \min(1, \exp(-T_0^{g-h}))$, which is integrable. By dominated convergence, $\int f_{T,h} \to \operatorname{vol} Q_h > 0$, since Q_h has interior when $h < h^*$. Scheffé's lemma then gives convergence in total variation. (2) Choose $\delta > 0$ with $\{g \geq h^* - 2\delta\} \subseteq U$. Put $s = h - h^* \geq 0$. Off U, $T^{g-h} \geq T^{-(s+2\delta)}$, so $\int_{U^c} f_{T,h} \leq C\exp(-\frac{1}{2}T^{-(s+2\delta)})$. Near a maximiser u_0, on the ball of radius $1/(ML)$, where $M = \max|v_k|$, every $\ell_k \geq h^* - 1/L$. So $f_{T,h} \geq \exp(-me\,T^{-s})$ there, and the mass of the ball is at least $cL^{-n}\exp(-me\,T^{-s})$. The ratio of the two bounds tends to 0, because $T^{-2\delta} \to \infty$. (3) g is concave and piecewise linear, and its superdifferential at u is $\operatorname{conv}\{v_k : k \in K(u)\}$ [28, Theorem 23.8]. A point maximises a concave function iff 0 lies in its superdifferential. Candidates satisfy this, and so does u^* by theorem 3(2). This recovers, in our setting, the theorem of Judd and Rietsch that the tropical critical point maximises the tropicalisation [26, Thm. 1.2]. KB8 corroborates (1)–(3) on $\mathbb{CP}^2$ and on F_1 with $\alpha = \frac{1}{5}$ and $\frac{3}{5}$. □

So the first level of the tropical limit sees exactly the max–min set. That set is the first step of the iterated max–min by which Fukaya, Oh, Ohta and Ono locate a fibre [17], and of the recursive construction of Judd and Rietsch [26]. The selection of u^* inside it is a second-level question.

Example 9 (the levels on F_1)**.** For $\mathbb{CP}^2$, and for F_1 with $\alpha < \frac{1}{3}$, $h^* = \frac{1}{3}$ and $\operatorname{argmax} g$ is the barycentre. The signed fibre of example 6(a) has $g = \alpha < h^*$. For F_1 with $\alpha > \frac{1}{3}$, $h^* = \frac{1-\alpha}{2}$, and $\operatorname{argmax} g$ is the closed segment $S = \{u_2 = \frac{1-\alpha}{2},\ \frac{1-\alpha}{2} \leq u_1 \leq \alpha\}$. Its relative interior consists of the candidates, and its endpoints are maximisers that are not candidates.

(a) At $h = h^*$ the u_1-marginal of $\nu_{T,h}$ tends to the *uniform* law on $[\frac{1-\alpha}{2}, \alpha]$. Leading order gives the selected point no extra mass.

(b) On S, $\ell_1 = u_1$ and $\ell_3 = \frac{1+\alpha}{2} - u_1$. The next max–min level is $h_2 = \max_S \min(\ell_1, \ell_3) = \frac{1+\alpha}{4}$, attained at $u_1 = \frac{1+\alpha}{4} = u_1^*$. For $h^* < h < h_2$, the u_1-marginal tends to the uniform law on $[h, \frac{1+\alpha}{2} - h]$. For $h > h_2$, it concentrates at u_1^*.

Sketch. Near S, the transverse factor $\exp(-T^{u_2-h} - T^{1-\alpha-u_2-h})$ does not depend on u_1. The coupling of u_2 to ℓ_3 matters only in boundary layers of width $O(1/L)$. So the u_1-marginal behaves as in theorem 8, now for $\min(\ell_1, \ell_3)$ on S. KB8 corroborates (a) and (b) for $\alpha = \frac{3}{5}$, at $h = \frac{1}{5}, \frac{3}{10}, \frac{9}{20}$ (fig. 1(c)). That the mean at level h^* equals u_1^* here is a symmetry of this example: u_1^* is the midpoint of S. The mechanism of selection is the second level, not the mean.

6 Soft evidence moves λ

This section starts the programme of conditioning existing structures on data and recording what changes. It treats one kind of data, soft evidence whose strength grows with the amount of data. Hard evidence (zero cells) and forgetting are not treated here.

Definition 10 (the data action)**.** Fix a reference scale N and put $T = e^{-N}$. Evidence that multiplies term k of W_T by $e^{-n\rho_k}$ after n observations, with rates $\rho_k \geq 0$, multiplies it by $T^{(n/N)\rho_k}$. So it replaces λ by

$$\lambda^{\text{eff}} = \lambda - s\rho, \qquad s = n/N.$$

Which term each observation weighs, and the scale N, are modelling choices. A likelihood of fixed size has $s \to 0$ as $N \to \infty$. It changes the coefficients smoothly, as in proposition 1, and changes no valuation.

Let Δ_m be the laws on the m facets, and let $B = \{q \in \Delta_m : \sum_k q_k v_k = 0\}$ be the *balance polytope*, the laws with mean exponent 0. It has dimension $m - n - 1$, because 0 lies in the interior of $\operatorname{conv}\{v_k\}$. Write $g_\lambda = \min_k \ell_k$, $h^*(\lambda) = \max g_\lambda$, and $F(\lambda)$ for the face of B on which $\langle q, \lambda\rangle$ is maximal.

Proposition 11 (the max–min level is a balance programme)**.** *(1) $h^*(\lambda) = \min_{q\in B}\big(-\langle q, \lambda\rangle\big)$. So h^* is concave and piecewise linear in λ.*

(2) Let K_0 be the union of the supports of the laws in $F(\lambda)$, and $\sigma_0 = \operatorname{conv}\{v_k : k \in K_0\}$. Then $K(u) = K_0$ on the relative interior of $\operatorname{argmax} g_\lambda$, and $K(u) \supsetneq K_0$ on its relative boundary. The candidates (theorem 8(3)) are exactly the relative interior, and $\dim \operatorname{argmax} g_\lambda = n - \dim \sigma_0$.

(3) The limit law of the positive critical point (theorem 3(2)) lies in $F(\lambda)$.

Proof. (1) $\max_u \min_k \ell_k = \max_u \min_{q\in\Delta_m} \sum_k q_k \ell_k(u)$, and the minimax theorem exchanges the two. For fixed q, $\sup_u \sum_k q_k(\langle v_k, u\rangle - \lambda_k)$ is $-\langle q, \lambda\rangle$ if $\sum_k q_k v_k = 0$ and $+\infty$ otherwise. (2) By complementary slackness, $q_k > 0$ for an optimal q forces $\ell_k(u) = h^*$ at every maximiser u, so $K(u) \supseteq K_0$ on $\operatorname{argmax} g_\lambda$. By strict complementarity [29], some maximiser has $\ell_j > h^*$ for every $j \notin K_0$. The same then holds on the relative interior of $\operatorname{argmax} g_\lambda$, which is $\{u : \ell_k(u) = h^* \ (k \in K_0),\ \ell_j(u) > h^* \ (j \notin K_0)\}$. A law in the relative interior of $F(\lambda)$ is positive on K_0 and has mean 0, so those points are candidates (lemma 2). At a boundary point, suppose 0 were in the relative interior of $\operatorname{conv}\{v_k : k \in K(u)\}$. Then there would be a balanced law positive on all of $K(u)$. That law would be optimal, since $\sum_k q_k \ell_k(u) = h^*$, so its support would lie in K_0, a contradiction. The linear span of $\{v_k : k \in K_0\}$ is its affine span, since it contains 0, so the equations $\ell_k = h^*$ on K_0 cut out dimension $n - \dim \sigma_0$. (3) The limit law q is balanced and supported on $K(u^*)$, where every $\ell_k(u^*) = h^*$. So $-\langle q, \lambda\rangle = \sum_k q_k \ell_k(u^*) = h^*$, and q is optimal. DW2 and DW3 corroborate (1)–(3) on 40 and 23 configurations. □

So candidates against selection appears twice, in dual spaces. In u-space, u^* is selected inside the relative interior of $\operatorname{argmax} g_\lambda$, which has dimension $n - \dim \sigma_0$. In law space, the limit law is selected inside $F(\lambda)$. The two can be non-trivial at different parameters (example 14).

Corollary 12 (where data can act)**.** *(1) The leading set K_0, the candidate dimension and the face F change only on the walls of the normal fan of B, the* outcome walls*. Within a chamber, h^* is affine in λ, and so is u^* when σ_0 is an n-simplex.*

(2) Evidence along a translation, $\rho_k = \langle v_k, t\rangle$, changes nothing but $u^* \mapsto u^* - st$. Evidence along the constant, $\rho = \mathbf{1}$, changes nothing but $h^* \mapsto h^* + s$. Only the quotient of $\mathbb{R}^m$ by these $n+1$ directions can cross a wall. It has dimension $m - n - 1$, the number of independent affine dependences (circuits) of $\{v_k\}$.

(3) Suppose there is one circuit a, normalised to be primitive and integral. The wall is $c(\lambda) = -\langle a, \lambda\rangle = 0$, and along $\lambda_0 - s\rho$ it is reached at $s^* = -c(\lambda_0)/\langle a, \rho\rangle$ when this is positive, never otherwise. Over rates with $\max_k \rho_k - \min_k \rho_k \le 1$, the least such s is $|c(\lambda_0)|/(\frac{1}{2}\|a\|_1)$.

Proof. (1) $F(\lambda)$ is constant on the relative interior of each cone of the normal fan of B; apply proposition 11. On a chamber, $h^* = -\langle q_0, \lambda\rangle$ for a fixed vertex q_0. When σ_0 is an n-simplex, u^* is the unique solution of $\ell_k(u) = h^*$, $k \in K_0$, which is linear in λ. (2) Under a translation, ℓ_k becomes $\ell_k(u + st)$. Under the constant, every ℓ_k shifts by s. Both are orthogonal to $B - B$, which consists of affine dependences. (3) B is a segment, and its normal fan has one wall, $\langle q_A - q_B, \lambda\rangle = 0$, where $q_A - q_B$ is a multiple of a. Since $\sum_k a_k = 0$, adding a constant to ρ leaves $\langle a, \rho\rangle$ unchanged. So the best value over range at most 1 is $\sum_k a_k^+ = \frac{1}{2}\|a\|_1$. □

Example 13 ($\mathbb{CP}^2$: no wall)**.** Here $m = n + 1$, so there is no circuit, and B is the single law $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. No soft evidence changes the leading set of $\mathbb{CP}^2$. Data only move the selected point and the level (DW4, DW7).

Example 14 (F_1 across its wall)**.** For general λ, put $L = -(\lambda_1 + \lambda_2 + \lambda_3)$ and $\alpha = -\lambda_1 - \lambda_3 + \lambda_4$. These are the size of the triangle and the length of the blown-up facet, and both are invariant under translation. The Kähler cone is $0 < \alpha < L$. Its walls are viability walls, where the polytope loses facet 4 or collapses, and X_Q changes. The one circuit is $a = (2, -1, 2, -3)$, with $c(\lambda) = 3\alpha - L$. The balance polytope is the segment from $q_A = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0)$ to $q_B = (0, \frac{1}{2}, 0, \frac{1}{2})$, and $h^* = \min\big(L/3, (L - \alpha)/2\big)$.

(a) The outcome wall $\alpha = L/3$ lies *inside* the Kähler cone: the manifold does not change there. Below it, $K_0 = \{1, 2, 3\}$ and the candidate is a point. Above it, $K_0 = \{2, 4\}$, the candidates form an open segment, and u^* is selected at the second level (example 9). The lower diagonal of the subdivision flips from v_1v_3 to v_2v_4.

(b) u^* is continuous and piecewise affine across the wall (fig. 2(a)). What jumps is the leading set, the candidate dimension ($0 \to 1$) and the mechanism of selection (leading order → second level). The signed fibre over $(\alpha, 1 - 2\alpha)$ reaches the barycentre at the wall. By the cited theorem [10], the number of torus fibres with non-vanishing Floer cohomology drops from two to one there. We claim nothing about the Fukaya category.

(c) *At the wall* ($\alpha = 1/3$, $L = 1$), the substitution $w = T^{1/6}x$ turns the quartic of example 6 into $x^4 - x - 1 = 0$ for every T. So the positive law is $(1/x, x^2, 1/x, 1/x^2)/\Sigma \approx (0.2156, 0.3922, 0.2156, 0.1766)$ for every T, where $x \approx 1.2207$. It is the interior point $(1 - t)q_A + tq_B$ of $F = B$, with $t \approx 0.353$. Here the candidate in u-space is a single point, but the law is selected inside a segment.

(d) The discriminant of $w^4 - T^{1/2}w - T^{1-\alpha}$ is $-27T^2 - 256T^{3-3\alpha}$, which never vanishes: no two critical points collide for $T > 0$. The outcome wall is where its two terms have equal order. It is a wall of the tropical discriminant, not of the discriminant itself.

(e) *Data.* Start at $\alpha = 1/5$, $L = 1$, so $c = -2/5$. Evidence against term 1, $\rho = e_1$, has $\langle a, \rho\rangle = 2$ and reaches the wall at $s^* = 1/5$, that is $n^* = N/5$ (fig. 2(b)). Along the way $\alpha/L = (1/5+s)/(1+s)$ stays in the Kähler cone. Evidence against term 4 ($\langle a, \rho\rangle = -3$) never reaches the wall. The most efficient rates with range at most 1 are $\rho = (1, 0, 1, 0)$, against terms 1 and 3, with $s = 1/10$.

DW4–DW7 corroborate (a)–(e).

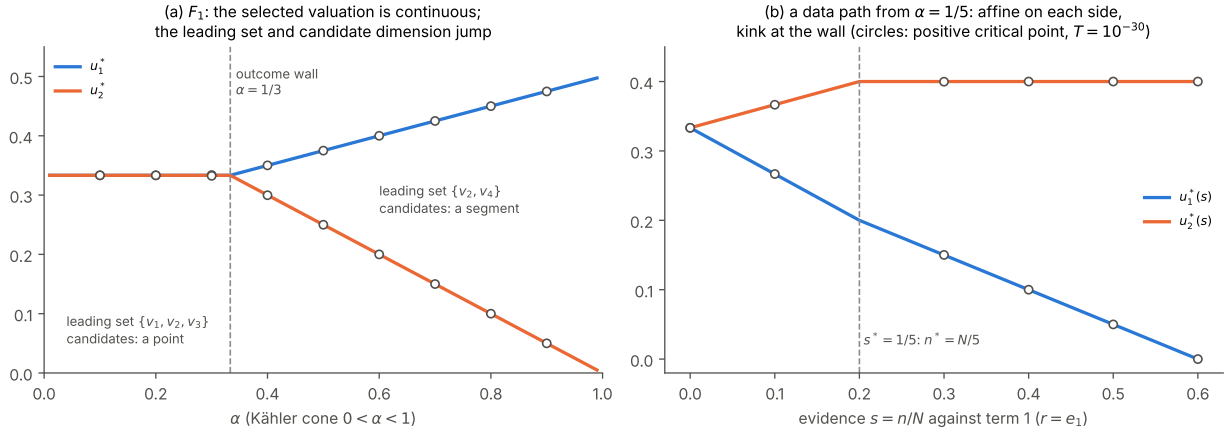


Figure 2: Soft evidence on F_1. (a) The selected valuation u^* across the Kähler cone. The curves are the chamber formulas and the circles the positive critical point at $T = 10^{-30}$. The outcome wall at $\alpha = 1/3$ changes the leading set and the candidate dimension, not the position of u^*; here $h^* = u_2^*$. (b) A data path from $\alpha = 1/5$ with evidence against term 1: $u^*(s)$ is affine on each side of $s^* = 1/5$, with a kink at the wall.

On F_1, then, the least data that changes the perspective changes *which terms carry the limit law*, not where the selected state sits. A decision that reads the leading set jumps at n^*. A decision that reads u^* continuously does not. Decision, contextuality and value-of-information walls need their own definitions, a decision model, a named complex and obstruction class, and an information-acquisition model, before they can be compared with the outcome walls. We record them separately and have not computed them.

Part II

Back to contingency tables

Part I went from toric geometry to conditioned probability. Part II runs the dictionary backwards, from a table of counts to the potential, and asks what the Floer-selected fibre means for the table. It works the independence model exactly. Every operation it uses, and every operation it does not yet treat, is recorded in the ledger (appendix A).

7 Two configurations of one model

An $r \times c$ table of counts t_{ij} has row margins t_{i+} and column margins t_{+j}. The independence model is the log-linear model whose design matrix A has one column per cell, the row indicator stacked on the column indicator. It is toric [12, 21]: its sufficient statistic is the vector of margins, and by Birch the MLE is the unique law in the model with the observed margins, $\hat{p}_{ij} = t_{i+}t_{+j}/n^2$ [7, 8]. The moment polytope of this model is $\Delta_{r-1} \times \Delta_{c-1}$, so its manifold is $\mathbb{CP}^{r-1} \times \mathbb{CP}^{c-1}$, which is Delzant and Fano.

The model carries two different point configurations, and they must not be confused.

Proposition 15 (cells and facets)**.** *(1)* Cells. *The rc columns of A satisfy $(r-1)(c-1)$ independent linear relations. These are spanned by the basic moves $+1$ at $(i,j),(k,l)$ and -1 at $(i,l),(k,j)$, which change a table without changing its margins. This is the Tier A support of proposition 1.*

(2) Facets. *The $r+c$ inward facet normals of $\Delta_{r-1}(L_1) \times \Delta_{c-1}(L_2)$ have exactly one circuit, $a = (c\,\mathbf{1}_r, -r\,\mathbf{1}_c)/\gcd(r,c)$. This is the configuration W is built from.*

Proof. (1) A has rank $r+c-1$, and the moves lie in its kernel and have rank $(r-1)(c-1)$ [12]. (2) $m - n - 1 = (r+c) - (r+c-2) - 1 = 1$. The row normals sum to 0, and so do the column normals. So $\sum_k a_k v_k = 0$, and $\sum_k a_k = rc - cr = 0$. CT1 checks both for $r, c \in \{2,3,4\}$. □

So the 2×3 model has two moves but one circuit. The moves act on tables with fixed margins, within a fibre. The circuit acts on λ, across Kähler classes. The second statement is a special case of a general count.

Proposition 16 (products add one circuit)**.** *If $v^{(1)}$ and $v^{(2)}$ both have 0 in the interior of their convex hulls, the product configuration has $(m_1 - n_1 - 1) + (m_2 - n_2 - 1) + 1$ independent circuits. The extra one is $(q^{(1)}, -q^{(2)})$ for balanced laws $q^{(i)}$ of the factors. Its wall is $h_1^* = h_2^*$ when each factor has a single balanced law.*

Proof. The count is $(m_1+m_2)-(n_1+n_2)-1$. The vector $(q^{(1)}, -q^{(2)})$ sums to $1-1=0$, and both halves balance. It is independent of the circuits of the factors, which sum to zero on each factor separately. When $B_i = \{q^{(i)}\}$, the balance programme gives $h^* = \min(h_1^*, h_2^*)$ (proposition 11), with a wall where they are equal. □

8 The independence model: its wall and its selection

In the coordinates $b =$ the first $r-1$ row margins and the first $c-1$ column margins, the facets of $Q = \Delta_{r-1}(L_1) \times \Delta_{c-1}(L_2)$ have $\ell_k(b)$ equal to the k-th margin. Here L_1 and L_2 are the totals of the row and column margins. So $W = W_{\text{row}}(y) + W_{\text{col}}(z)$ with $W_{\text{row}} = y_1 + \cdots + y_{r-1} + T^{L_1}/(y_1 \cdots y_{r-1})$, and similarly for columns.

Theorem 17 (the independence model)**.** *(1) $h^* = \min(L_1/r,\, L_2/c)$. The leading set is the row facets if $cL_1 < rL_2$, the column facets if $cL_1 > rL_2$, and all $r+c$ facets on the wall $cL_1 = rL_2$. The wall is $c(\lambda) = -\langle a, \lambda\rangle = (cL_1 - rL_2)/\gcd(r,c) = 0$.*

(2) The critical points of W are $y_i = \zeta T^{L_1/r}$, $z_j = \eta T^{L_2/c}$ with $\zeta^r = \eta^c = 1$, so there are rc of them. All have the valuation $u^* = (L_1/r, \ldots; L_2/c, \ldots)$, the uniform margins, for every L_1, L_2. The positive law is exactly proportional to $T^{L_1/r}$ on each row facet and $T^{L_2/c}$ on each column facet.

(3) *Off the wall the candidates have dimension* $c-1$ *(if* $cL_1 < rL_2$*) or* $r-1$*, and the uniform margins are selected at the second level. On the wall the candidate is the single point* u^**.*

(4) *By the cited theorem [10], exactly one torus fibre has non-vanishing Floer cohomology, for every* L_1, L_2*. So the outcome wall of the independence model changes no Floer data. On* F_1 *it does (example 14(b)).*

Proof. (1) Each factor has the single balanced law, uniform on its facets, with values L_1/r and L_2/c. Apply proposition 16. (2) The variables separate. $\partial W_{\text{row}}/\partial y_i = 0$ gives $y_i \prod_j y_j = T^{L_1}$ for every i, so all y_i are equal and $y^r = T^{L_1}$. The term values are then $T^{L_1/r}$ times a root of unity. (3) Off the wall the leading set is one factor's facets, so σ_0 has dimension $r-1$ (or $c-1$); apply proposition 11(2). (4) All critical points share one valuation. CT2 and CT3 check (1)–(3), with exact rationals and gradients. □

9 Margins against evidence

Data enter in two ways, and the note keeps them apart.

Proposition 18 (two roles of data)**.** *(1)* Observed margins b *pick a point of* Q*, the fibre* $\mu^{-1}(b)$ *and the MLE (Birch). For the independence model,* $\ell_k(b)$ *is exactly the* k*-th margin, the facet* $\{\ell_k = 0\}$ *is a zero margin, and the MLE lies in the open model iff every margin is positive [8]. For general hierarchical models the facets correspond to facial sets of cells, and the MLE fails to exist exactly when the margins lie on a proper face [14, 15]. We use only the independence case.*

(2) Evidence on the terms of W *moves* λ *(definition 10), and with it* u^**,* h^* *and the walls. With* λ *fixed,* u^* *does not depend on* b*.*

(3) *Both* b *and* u^* *are points of* Q*, in moment coordinates. The valuation of a critical point is the moment coordinate of its torus fibre* $L(u^*) = \mu^{-1}(u^*)$ *[10]. So the MLE fibre and the Floer-selected fibre can be compared in one space. For the independence model we use the Fisher–Rao distance within the model, between* $\hat{p}$ *and the law at* u^* *(uniform margins). With surveys of sizes* n_1 *for rows and* n_2 *for columns,*

$$d(\hat{p}, p^*)^2 = n_1\, d_\Delta(\hat{p}_{\text{row}}, \mathbf{1}/r)^2 + n_2\, d_\Delta(\hat{p}_{\text{col}}, \mathbf{1}/c)^2, \qquad d_\Delta(p,q) = 2\arccos \sum_i \sqrt{p_i q_i}.$$

Joint sampling of n *cells is the case* $n_1 = n_2 = n$*. A coarser index is the* centrality $g(b)/h^*$*: the smallest margin divided by its largest possible value.*

Proof. (1) The first statements are direct, and the MLE formula shows positivity. (2) is corollary 12 together with the fact that W does not involve b. (3) The Fisher information of the independence family is block diagonal, $n_1 \operatorname{Cov}(\text{row}) \oplus n_2 \operatorname{Cov}(\text{col})$, because the log-likelihood separates into a row part and a column part. So the model is a Riemannian product of two scaled simplices, and on each simplex the Fisher–Rao distance is $2\arccos$ of the Bhattacharyya coefficient [2]. CT4 and CT5 check (1) and the block structure. □

The MLE fibre and the Floer-selected fibre coincide exactly when the margins are uniform. That the Floer fibre is the uniform-margin table, the one farthest from every zero-margin degeneracy in lattice distance, is a reading of the cited theorem, not a new statistical result. The polytope used here is the *base*, the space of margins. Section 11 turns to the *fibre* over a point of it, the coupling polytope, where the same margins become the Kähler class.

10 Data and walls in the independence model

Example 19 (2×2 and 2×3)**.** Under joint sampling, $L_1 = L_2 = n$. The 2×2 table therefore sits on its wall for every n, which is the monotone $\mathbb{CP}^1 \times \mathbb{CP}^1$. For 2×3, $cL_1 - rL_2 = n > 0$, so $h^* = n/3$, the column facets lead, and the candidates are the open segment of row margins $b_1 \in (n/3, 2n/3)$ with uniform column margins. Its endpoints are maximisers but not candidates. The selected point is $b_1 = n/2$. To reach the wall the column survey must grow to $n_2 = 3n/2$: $n/2$ more column observations. In the terms of corollary 12(3), this is ρ on a column facet with $\langle a, \rho \rangle = -r/\gcd$, so $s^* = n/2$. Crossing it changes the leading set and the candidate dimension. By theorem 17(4) it changes no Floer data. CT6 checks every number.

Example 20 (hard evidence: a structural zero)**.** Declare cell $(2, 2)$ of the 2×2 table structurally zero. This is the quasi-independence model on three cells. The three remaining cells span a unimodular triangle, which is *not* a face of the square: hard evidence need not restrict to a toric divisor. Its facet configuration is that of $\mathbb{CP}^2$, which has no circuit, so the one wall of the 2×2 model is gone. CT7 checks this. The general case, deleting cells of larger tables, is open.

11 One coupling polytope through every layer

What this section shows, in plain words. Fix two margins, a row law f and a column law g. The tables (couplings) with those margins form a polygon, the *fibre*. That polygon is the picture of a toric surface, and everything in Part I can be drawn on it.

- The usual conditional law of a table given its margins, the one behind Fisher's exact test, *is* the toric Kähler structure (fig. 4).
- The positive critical point of W picks the *leximin* coupling: the one whose smallest cell is as large as possible (fig. 3).
- The positive thimble is a family of laws on couplings. It runs from "every coupling equally likely" to "only the leximin coupling" (fig. 5).
- Two kinds of wall cut the space of margins. Blow-up walls change the surface; leximin walls change which cells lead. Here one count of data crosses a leximin wall and changes the Floer fibres, while the surface stays the same (fig. 6).

The example is the one followed in the architecture note [5]: $f = (2/5, 3/5)$, $g = (3/10, 43/100, 27/100)$.

11.1 Base and fibre

A pair of margins is a point of the *base*, $\Delta_{n-1} \times \Delta_{m-1}$ (section 8). Over it sits the *fibre*, the coupling polytope $\mathrm{Cpl}(f, g)$ of tables with those margins. For nondegenerate margins the fibre is Delzant, its facets are cells $\{\alpha_c = 0\}$, and the Kähler class of its toric manifold is the product coupling, $[\omega_{f,g}] = \sum_c f_i g_j [D_c]$ [4]. So the same data play two roles:

	base (sections 7 to 9)	fibre (this section)
points	margins (f, g)	couplings α with those margins
facets	a zero margin	a zero cell
the margins are	a point (the Birch/MLE fibre)	the Kähler class: λ_c is read off (f, g)
W is a sum over	margin facets	cell facets
Tier A support	cells, with $(r-1)(c-1)$ moves	integer tables with these margins

For 2×3, write a coupling by $x = \alpha_{11}$, $y = \alpha_{12}$. Then $\alpha_{13} = f_1 - x - y$, $\alpha_{21} = g_1 - x$, $\alpha_{22} = g_2 - y$ and $\alpha_{23} = x + y - (f_1 + g_1 + g_2 - 1)$, so cell c has $\ell_c = \langle v_c, (x, y) \rangle - \lambda_c$ with $v_c \in \{(1,0), (0,1), (-1,-1), (-1,0), (0,-1), (1,1)\}$.

Example 21 (the fibre)**.** For $f = (2/5, 3/5)$, $g = (3/10, 43/100, 27/100)$ the fibre is the pentagon with vertices $(13/100, 0)$, $(3/10, 0)$, $(3/10, 1/10)$, $(0, 2/5)$, $(0, 13/100)$ and area $1331/20000$, so $X = \mathrm{Bl}_2\mathbb{CP}^2$. Its facets are the cells $11, 12, 13, 21, 23$. Cell 22 never vanishes, since $\alpha_{22} \geq f_2 + g_2 - 1 = 3/100$. Scaled by $n = 100$, its lattice points are the 715 integer tables with margins $(40, 60)$ and $(30, 43, 27)$. By Pick's theorem, $715 = 665.5 + 97/2 + 1$. CF1 and CF2 check this exactly.

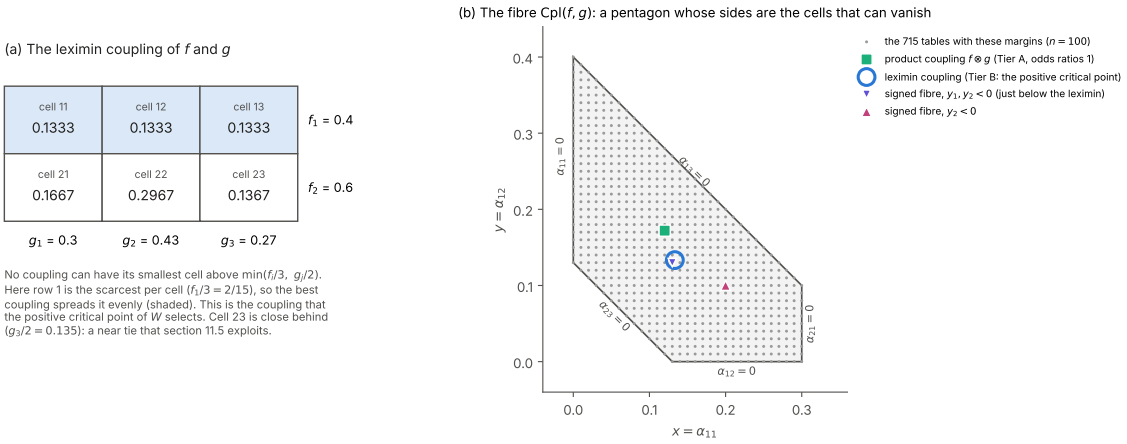


Figure 3: The example at a glance. (a) The leximin coupling: row 1 is the scarcest per cell, so it is spread evenly, and every other cell is larger. (b) The fibre, with the 715 tables as dots. Each side is a cell that can vanish. Marked: the product coupling, selected by Tier A; the leximin coupling, selected by the positive critical point of W; and the two real critical points with a negative coordinate (signed fibres).

11.2 Tier A on the fibre is Fisher's noncentral hypergeometric family

Proposition 22 (the conditional law of a table)**.** *Let S be the integer tables with margins Nf, Ng, written by (t_{11}, t_{12}), with weights $c_t = 1/\prod_{ij} t_{ij}!$.*

(1) For any cell parameters $\theta_{ij} > 0$, the law of a table under independent Poisson (or multinomial) sampling with means $\propto \theta_{ij}$, conditioned on its margins, is the monomial law $q(u) \propto c_t\, e^{\langle u, (t_{11}, t_{12}) \rangle}$. Here $u_1 = \log \frac{\theta_{11}\theta_{23}}{\theta_{13}\theta_{21}}$ and $u_2 = \log \frac{\theta_{12}\theta_{23}}{\theta_{13}\theta_{22}}$ are the two log odds ratios. This is Fisher's noncentral hypergeometric family. At $u = 0$ it is the central hypergeometric law, the null law of Fisher's exact test [16].

(2) By proposition 1, $\varphi(u) = \log \sum_t c_t e^{\langle u, t \rangle}$ is a Kähler potential. The moment map is the expected table, and the Kähler metric is the Fisher information of the odds ratios. Since S is the set of lattice points of the Delzant lattice polygon $N \cdot \mathrm{Cpl}(f, g)$, the form extends to all of $X = \mathrm{Bl}_2\mathbb{CP}^2$.

(3) At $u = 0$ the mean is exactly $N\, f \otimes g$: Tier A at odds ratios 1 selects the product coupling.

(4) Every interior coupling is the mean for exactly one u (Birch). In the example, the leximin coupling needs odds ratios $(e^{u_1}, e^{u_2}) \approx (0.82, 0.46)$.

Proof. (1) $\prod_{ij} \theta_{ij}^{t_{ij}}/t_{ij}!$ restricted to fixed margins depends on t only through c_t and $t_{11}u_1 + t_{12}u_2$. Substituting $t_{13}, t_{21}, t_{22}, t_{23}$ from the margins gives the two log odds ratios. (2) is proposition 1 and its compact case. (3) The central hypergeometric mean is $t_{i+}t_{+j}/N$ [8]. (4) is proposition 1(3). CF3 checks (1) and (3), the latter in exact rationals, and computes (4). □

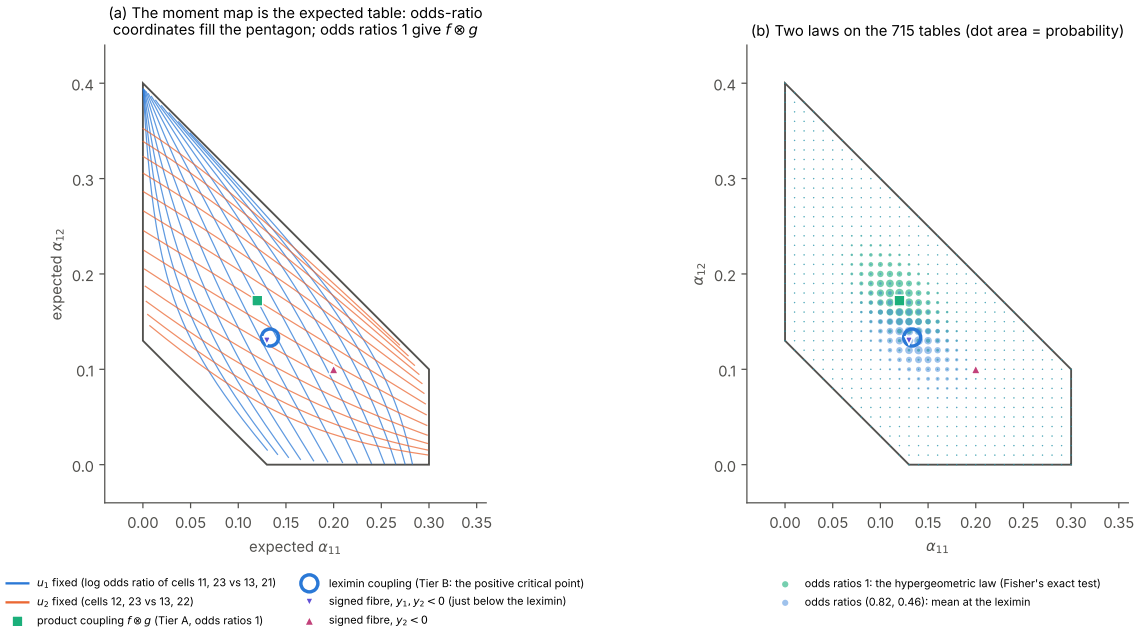


Figure 4: Tier A. (a) The moment map sends the plane of log odds ratios onto the open pentagon. Blue lines hold u_1 fixed and orange lines hold u_2 fixed. (b) Two laws on the 715 tables: the hypergeometric law (odds ratios 1, centred on $f \otimes g$), and the law with odds ratios $(0.82, 0.46)$, centred on the leximin coupling.

11.3 Tier B: the selected coupling is the leximin one

Theorem 23 (the max–min coupling)**.** *Let (f, g) be nondegenerate $n \times m$ margins, and let W be the potential of* $\mathrm{Cpl}(f, g)$*, a sum over its facet cells.*

(1) $h^* = \max_\alpha \min_{c\ facet} \alpha_c = \min\bigl(\min_i f_i/m,\ \min_j g_j/n\bigr)$.

(2) Every cell of a binding *row or column, one attaining the minimum, is a facet. If exactly one line binds, the leading set K_0 is that line's cells.*

(3) The candidate set has dimension $(n-2)(m-1)$ for a binding row and $(n-1)(m-2)$ for a binding column. For 2×3, a binding row gives a single candidate, so leading order selects. A binding column gives a segment, and the next level selects.

(4) The positive critical point of W has its valuation in the candidate set (theorem 3 and proposition 11). For 2×3 it sits over the leximin coupling. This is checked on random margins, not proved; it is the recursion of [17, 26] on this family.

Proof. (1) A row of m cells sums to f_i, so its smallest cell is at most f_i/m; columns are the same. Conversely, put h (the claimed value) in every cell and add the product of the residual margins $f_i - mh \geq 0$ and $g_j - nh \geq 0$, whose totals agree. This gives a coupling with every cell at least h. (2) A cell with $\alpha_c > 0$ on all of Cpl has $f_i + g_j > 1$. For a binding row i that would give $f_i > \sum_{k \neq j} g_k \geq (m-1)\min_k g_k$ and $f_i \leq (m/n)\min_k g_k$, so $n(m-1) < m$, which is impossible. If row i alone binds, the residual margins of the other rows and of all columns are positive, so the coupling of (1) has every other cell strictly above h^*. The row's cells, which average h^*, equal it at every maximiser. (3) The m cell functionals of a row sum to the constant f_i, so their normals span $m-1$ dimensions and contain 0 in the relative interior of their hull. Apply proposition 11(2) in dimension $(n-1)(m-1)$. (4) CF6 checks it on 20 random rational margins at $T = 10^{-60}$. CF4 checks (1)–(3) on about 1000 random margins of shapes 2×3, 3×3, 2×4 and 3×4, against the facet-only programme. □

In words: *the selected coupling spreads the scarcest row or column evenly over its cells.* In the example, row 1 binds with $f_1/3 = 2/15$. So $h^* = 2/15$, $K_0 = \{11, 12, 13\}$, and the positive critical point sits over the leximin coupling $(2/15, 2/15)$. The product coupling $f \otimes g = (0.12, 0.172)$ has the single leading cell 13, at $0.108 < h^*$, so it is not a candidate. That is the Tier-B reason why "the product coupling carries no brane" [3].

Example 24 (the five critical points)**.** $W = y_1 + y_2 + T^{2/5}/(y_1 y_2) + T^{3/10}/y_1 + T^{-13/100} y_1 y_2$ has 5 critical points, twice the area $5/2$ of its Newton polygon (Kushnirenko). Exact resultants at $T = 10^{-100}$ and 10^{-200} place them as follows.

- Three lie over the leximin coupling, one of them positive.
- One is real with $y_1, y_2 < 0$, over $(13/100, 13/100)$. Its leading cells are $\{11, 12, 23\}$, whose normals have the signed balance $(1,0) + (0,1) - (1,1) = 0$ but no positive one.
- One is real with $y_2 < 0$, over $(1/5, 1/10)$. Its leading cells are $\{12, 13, 21\}$, also with a signed balance only.

This is theorem 3(4) on the example. By the cited theorem [10], three torus fibres have non-vanishing Floer cohomology. This agrees with [3, Thm. 8.2]. CF5 checks the count, the positions, the signs and the leading sets.

11.4 The positive thimble on couplings

By theorem 8, the positive thimble with $\hbar = T^h$, read in $u = \log y / \log T$, tends to the uniform law on $Q_h = \{\alpha : \text{every facet cell} \geq h\}$ for $h < h^*$, and concentrates on argmax at h^*. On the fibre this has a plain reading. At $h = 0$ it is the uniform law on all couplings, which is Duistermaat–Heckman, and the limit of the uniform law on tables. As h grows it is uniform on the couplings whose every cell is at least h. At $h^* = 2/15$ it is the leximin coupling (fig. 5; CF7). The thimble is thus a one-parameter family of probability laws on couplings, from "all equally likely" to "the fairest one".

11.5 Two kinds of wall, and one count of data

In the base, the margins (f, g) move the Kähler class of the fibre. Two families of walls cut the base.

- *Kähler-chamber walls* $f(I) = g(J)$. Across them the polygon gains or loses a side, which is a toric blow-up or blow-down [4]. These are the viability walls of the ledger.

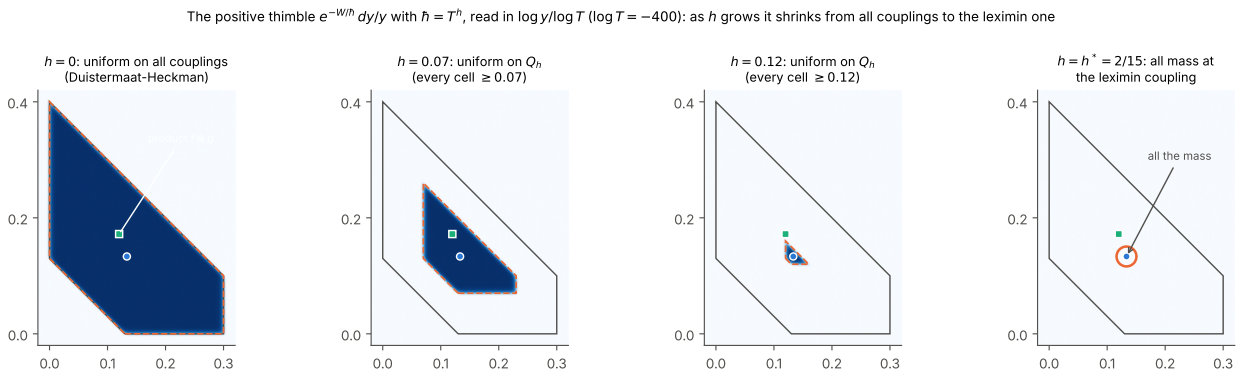


Figure 5: The positive thimble on the fibre, at four values of h, with $\log T = -400$. Dashed: the shrunk polygon Q_h, every cell at least h. The law is uniform on Q_h, and at $h^* = 2/15$ all its mass is at the leximin coupling (circle). The product coupling (square) is passed over.

- *Leximin walls*, where two of the ratios f_i/m, g_j/n tie. Across them the binding line, and with it the leading cells and the candidate dimension, jumps (theorem 23). These are the outcome walls of corollary 12, here explicit.

They are different walls (fig. 6(a)). The monotone point, with uniform margins, lies on all of them.

Example 25 (one count)**.** In the example, $f_1/3 = 2/15$ and $g_3/2 = 27/200$ differ by only $1/600$. Move one observation of 100 from column 3 to column 2, so that $g = (30, 44, 26)/100$. Then:

- column 3 binds, $h^* = 13/100$, and the leading cells are $\{13, 23\}$;
- the candidates become a segment, and the leximin coupling, selected at the next level, is $(27/200, 27/200)$;
- the real critical point with $y_1, y_2 < 0$ moves onto the leximin fibre, so the distinct valuations drop from three to two. By the cited theorem, so does the number of torus fibres with non-vanishing Floer cohomology.

The surface does not change: the nearest blow-up wall, $f_1 = g_2$, is three counts away. So in this example the least data that changes the perspective is one count, and it changes the leading cells and the Floer fibres, not the manifold (fig. 6(b); CF8, with exact resultants at $T = 10^{-100}$ and 10^{-300}).

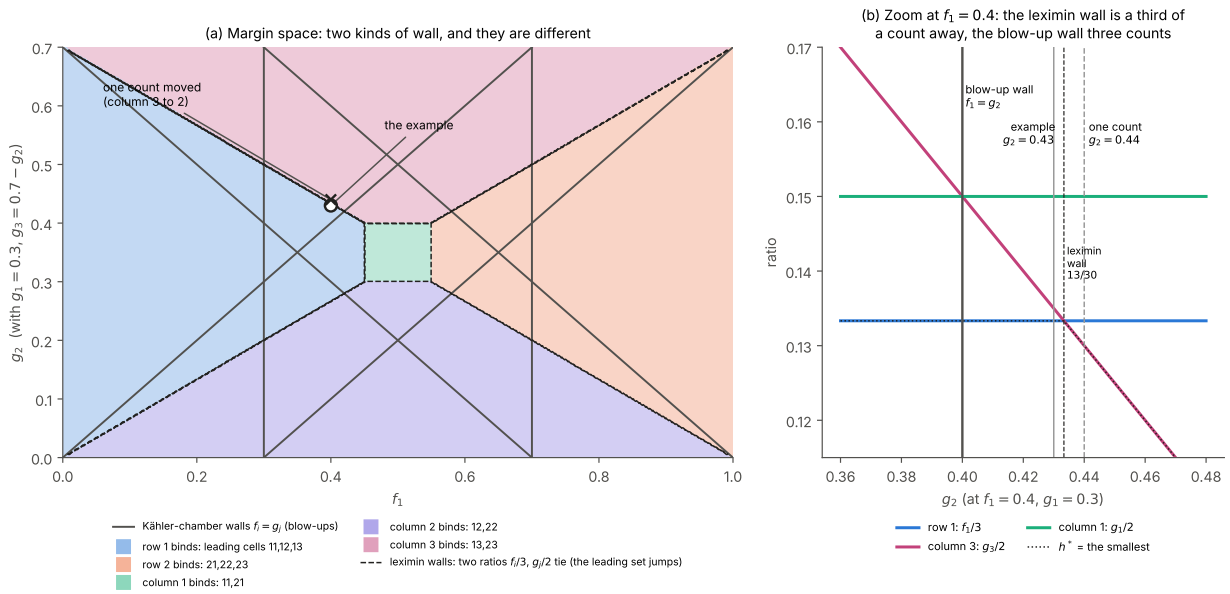


Figure 6: Margin space, in the slice $g_1 = 0.3$. (a) Colours show which row or column binds. Dashed lines are the leximin walls and solid lines the blow-up walls $f_i = g_j$. The example (circle) sits almost on a leximin wall, and one count (cross) crosses it. (b) Along $f_1 = 0.4$: the three ratios $f_1/3$, $g_3/2$, $g_1/2$, whose minimum is h^*. The leximin wall is a third of a count from the example, while the blow-up wall $f_1 = g_2$ is three counts away.

11.6 The example through every layer

layer	in the example	where
fibre	pentagon, area 1331/20000, facets $11, 12, 13, 21, 23$; 715 tables (Pick)	example 21; CF1–2
Tier A	Fisher's noncentral hypergeometric family; odds ratios 1 give $f \otimes g = (0.12, 0.172)$; the leximin needs $(0.82, 0.46)$	proposition 22; CF3
Tier B	$h^* = f_1/3 = 2/15$; row 1 binds; one candidate, the leximin coupling $(2/15, 2/15)$	theorem 23; CF4, CF6
sign split	five critical points; signed fibres over $(13/100, 13/100)$ and $(1/5, 1/10)$	example 24; CF5
Floer (cited)	three fibres with non-vanishing Floer cohomology; none over $f \otimes g$	[10, 3]
thimble	uniform on couplings at $h = 0$; uniform on Q_h; the leximin coupling at h^*	theorem 8; CF7
walls	leximin wall $1/600$ away in ratio; blow-up wall $f_1 = g_2$ three counts away	fig. 6
one count	column 3 binds; candidates a segment; three Floer fibres become two; same surface	example 25; CF8
not claimed	other thimbles, Stokes data, stability conditions, the category	section 12

12 Outlook

Five questions are left open.

- *Novikov weights and environments.* Section 6 makes one direction precise: soft evidence of size n on the scale N shifts λ by $-(n/N)\rho$. Whether the environment orders s^{r_e} of the layers paper act in the same way, and how hard evidence and forgetting act, is open.
- *The other walls.* Decision, contextuality and value-of-information walls (section 6), and the finite-n sharpness of the transition at an outcome wall, measured in the Fisher metric.
- *Beyond Fano.* For non-Fano toric manifolds, W acquires corrections [17]. Theorem 3 is about the leading potential only.
- *Signed branches and Stokes data.* The non-positive critical points have their own thimbles, which are not real and carry no probability law. How the positive thimble decomposes against them, as $\hbar$ and T vary, is Picard–Lefschetz and Stokes data. We do not compute it here. In particular, we do not know whether the signed fibre of example 6(a) has a probabilistic shadow.
- *Wall-crossing.* The layers paper resolves degenerate directions by blow-ups off the torus orbits. Whether these correspond to wall-crossing in torus fibrations is a separate question, which needs geometric data that the finite setting does not supply.

A The ledger

`LEDGER.yaml` at the root of the repository is the single source for this table, which `kahler/render_ledger.py` generates. Status tags: [P] proved here, [U] used (classical), [C] cited and not proved, [E] exact computation, [N] numerical only, [D] a definition, [O] open. The category level is never filled by assumption. For each experiment the ledger also records the thresholds $s_{\text{prob}}, s_{\text{support}}, s_{\text{trop}}, s_{\text{crit}}, s_{\text{Stokes}}, s_{\text{context}}$, each as a number, "never" or "undefined". For F_1 along $\rho = e_1$ from $\alpha = 1/5$ they are: $s_{\text{trop}} = 1/5$; s_{crit} never; s_{support} never; s_{prob}, s_{Stokes} and s_{context} undefined until a dominance notion, a phase of $\hbar$, and a complex with an obstruction class are chosen.

operation	contingency / toric	mirror (W, λ)	Floer (cited); category	wall certificate; checks
soft-translation: Soft evidence along a translation, rho_k = ⟨v_k, t⟩ (an exponential tilt)	[D] reweighting by a log-linear factor: a reparametrisation of the same model; [P] Q translated by -s t; same Kahler class	[P] u* → u* - s t; h*, leading set and candidates unchanged	[C] the same fibres, translated; category [O] not addressed	none: rho is orthogonal to B - B (Corollary 12(2)); CHECKS DW7
soft-constant: Soft evidence along the constant, rho = 1 (normalisation)	[D] a common factor: removed by normalisation; [P] every facet moves out by s	[P] h* → h* + s; nothing else changes	[C] no change in which fibres carry Floer cohomology; category [O] not addressed	none: sum_k a_k = 0 for every affine dependence a; CHECKS DW7
soft-circuit: Soft evidence with a circuit component (lam_eff = lam - (n/N) rho)	[E] independence: a larger row or column survey (L1 = n1, L2 = n2); 2x3 needs n/2 more column observations to reach its wall; [P] moves the Kahler class; viability walls are the walls of the Kahler cone (F_1: alpha/L in 0, 1)	[P] outcome walls = normal fan of the balance polytope B; leading set, candidate dimension and face F jump there; h* concave piecewise linear	[C] F_1: two fibres with HF ≠ 0 below alpha/L = 1/3, one above (Cho-Oh); independence: one fibre on both sides; category [O] no categorical wall-crossing claimed	one circuit a: s* = -c(lam_0) / ⟨a, rho⟩; least s over rate range ≤ 1 is |c| / (||a||_1 / 2). F_1: a = (2,-1,2,-3), c = 3 alpha - L; independence: a = (c 1_r, -r 1_c)/gcd, c(lam) = (c L1 - r L2)/gcd; CHECKS DW2, DW4, DW5, DW6, DW7, CT2, CT6

operation	contingency / toric	mirror (W, λ)	Floer (cited); category	wall certificate; checks
margins: Observed counts (margins b)	[U] the MLE has the observed margins (Birch); independence: rows x cols / n^2; [U] the fibre over b; l_k(b) is the k-th margin (independence: exact)	[P] u* does not depend on b while lam is fixed	[D] compare L(b) with the Floer-selected L(u*), in Fisher-Rao distance within the model; category [O] not addressed	support: min_k l_k(b) = 0; centrality g(b)/h* = smallest margin / (n / max(r, c)) for independence with joint sampling; CHECKS CT4, CT5
margins-on-the-fibre: Margins (f, g) read on the fibre, the coupling polytope Cpl(f, g)	[E] the fibre's lattice points are the tables with these margins; Tier A is Fisher's noncentral hypergeometric family, odds ratios 1 at f(x)g; [U] Cpl(f, g) is Delzant; class = product coupling; 2x3: CP^2, CP^1 x CP^1, F_1, Bl_2 CP^2, dP_6 (coupling-polytopes paper)	[P] h* = min(f_i/m, g_j/n); leading cells = the binding row or column; candidate dimension (n-2)(m-1) or (n-1)(m-2)	[C] example: three fibres with HF $\neq 0$; one count moved: three become two (valuations by exact resultants); category [O] not addressed	the argmin of f_i/m, g_j/n changes. Example: f1/3 = 2/15 vs g3/2 = 27/200 (1/600 apart): one count crosses; the blow-up wall f1 = g2 is three counts away; CHECKS CF1, CF2, CF3, CF4, CF5, CF6, CF7, CF8
refinement: Refinement: add a distinction (a facet normal, a toric blow-up)	[O] which table operation adds a facet is not yet identified; [U] CP^2 $\to$ F_1 is the blow-up at a point	[E] creates the circuit (2,-1,2,-3) and the wall alpha/L = 1/3; CP^2 had none	[C] F_1 gains a second fibre with HF $\neq 0$ below the wall; category [O] not addressed	m - n - 1 increases by one; CHECKS DW4
structural-zero: Hard evidence: a structural zero (support restriction)	[E] 2x2 with one structural zero: quasi-independence on three cells; [E] the unit square becomes a unimodular triangle, not a face of it: CP^1 x CP^1 is replaced by CP^2	[E] the one circuit and the one wall disappear	[C] CP^2: one fibre (the Clifford torus); category [O] not addressed	face test fails; DW circuits of the triangle: none; CHECKS CT7
independent-product: Independent product of two models	[E] independence of rows and columns: Delta_(r-1) x Delta_(c-1); [U] the product manifold	[P] adds exactly one circuit, the relative scale (q1, -q2); u* = (u1*, u2*) never moves	[C] independence: exactly one fibre with HF $\neq 0$ for every L1, L2, so this wall changes no Floer data; category [O] Kunneth-type statements not addressed	m - n - 1 = (m1 - n1 - 1) + (m2 - n2 - 1) + 1; CHECKS CT1, CT2, CT3
forgetting: Forgetting an environment / marginalisation	[P] the layers paper: the forgetting square, reversal walls, pooled orders; [O] the image configuration under a linear map of exponents: not yet worked out	[O] not yet worked out	[O] not addressed; category [O] not addressed	layers paper, theorem "contextual Bayes with an environment"; CHECKS EN1, EN2, EN3
coupling: Dependent coupling, convolution, intervention, Markov evolution	[O] not yet worked out; [O] convolution would be a Minkowski sum of supports	[O] not yet worked out	[O] not addressed; category [O] not addressed	-; CHECKS –

Checks

code	`engine/pce/kahlerbridge.py`, `engine/pce/datawalls.py`, `engine/pce/contingency_toric.py`, `engine/pce/couplingfibre.py`
checks	`engine/check_kahlerbridge.py` (KB, 8/8), `engine/check_datawalls.py` (DW, 7/7), `engine/check_contingency_toric.py` (CT, 7/7), `engine/check_couplingfibre.py` (CF, 8/8), `engine/check_ledger.py` (LG, 3/3)
figures	`kahler/fig_kahler.py`, `kahler/fig_datawalls.py`, `kahler/fig_coupling.py`
ledger	`LEDGER.yaml`, rendered by `kahler/render_ledger.py`

The checks corroborate the statements numerically, or by exact rational arithmetic where marked; they do not prove them. The proofs stand on their own. The F_1 blow-up sizes used are $\alpha = 1/5, 1/4, 2/5, 3/5, 4/5$.

KB1	proposition 1(1)–(2): moment map $= \nabla\varphi$, Fisher information $= \operatorname{Hess}\varphi$, positive definite (30 supports)
KB2	proposition 1(4): $\psi = \sum q \log(q/c)$, $\nabla\psi = u(p)$, $\operatorname{Hess}\psi = \text{Fisher}^{-1}$ (20 supports)
KB3	proposition 1(5): $\int \det \operatorname{Hess}\varphi$ over boxes equals the area of the moment image
KB4	theorem 3(1)–(2): one positive critical point, mean-zero law, the valuations of examples 5 and 6, balanced leading terms
KB5	theorem 3(3): the segment of candidates for $\alpha = 3/5$, and its endpoints $(1/5, 1/5)$, $(3/5, 1/5)$, with leading sets $\{v_1, v_2, v_4\}$, $\{v_2, v_3, v_4\}$ and no positive balance; the barycentre alone for $\alpha = 1/5$ and for $\mathbb{CP}^2$ (exact rational points)
KB6	theorem 3(4): all four critical points of F_1 in 80-digit arithmetic; the signed critical point and its signed balance
KB7	proposition 7: Hessian $\operatorname{diag}(H, -H)$ at y_T; Laplace covariance $\hbar H^{-1}$ with relative error $O(\hbar/W_T(y_T))$ ($\mathbb{CP}^2$, F_1 with $\alpha = 3/5$)
KB8	theorem 8, example 9: moments against the uniform law on Q_h for $h < h^*$; concentration on $\operatorname{argmax} g$ at h^*; the uniform segment; the levels 3/10, 9/20; $\log T = -200, -800$
DW1	definition 10: evidence $e^{-n\rho}$ at $T = e^{-N}$ is the shift $\lambda - (n/N)\rho$
DW2	proposition 11(1)–(2): duality; optimal support = leading set at the centre of argmax; $\dim \operatorname{argmax} = n - \dim \sigma_0$ (40 random configurations)
DW3	proposition 11(3): the limit law lies in the optimal face (23 cases)
DW4	example 14(a): the circuit $(2, -1, 2, -3)$; the wall $\alpha/L = 1/3$ inside $0 < \alpha/L < 1$; the diagonal flip (exact rationals); $\mathbb{CP}^2$ one chamber
DW5	example 14(b),(c): candidates and valuations for $\alpha = 1/10, \ldots, 9/10$; the wall law from $x^4 = x + 1$ at $T = 10^{-20}, 10^{-60}$
DW6	example 14(d): the discriminant, symbolically; its terms tie only at $\alpha = 1/3$
DW7	corollary 12, example 14(e): $s^* = 1/5$ for $\rho = e_1$; no wall for e_4, $(0, -1, 0, 1)$, translations or constants; least evidence 1/10; $\mathbb{CP}^2$ never
CT1	proposition 15: $(r-1)(c-1)$ moves; one facet circuit ($r, c \in \{2, 3, 4\}$)
CT2	theorem 17(1): h^*, the leading sets and the wall $cL_1 = rL_2$ (exact rationals)
CT3	theorem 17(2)–(3): the rc critical points and their common valuation; the exact positive law; candidate dimensions
CT4	proposition 18(1): $\ell_k(b) =$ margins; MLE margins and positivity (200 tables)
CT5	proposition 18(3): the block-diagonal Fisher information; properties of d_Δ
CT6	example 19: $h^* = n/3$, the candidate segment, u^*, and $n/2$ more column observations
CT7	example 20: the triangle is not a face; no circuit
CF1–CF2	example 21: the pentagon, its facets, Delzant; 715 tables (Pick), exact
CF3	proposition 22: odds-ratio identity; exact hypergeometric mean $(12, 86/5)$; Birch at the leximin
CF4	theorem 23(1)–(3): the closed form against the facet programme; binding cells are facets; candidate dimensions
CF5	example 24: five critical points by exact resultants; positions, signs, leading sets, signed balances
CF6	theorem 23(4): positive valuation = leximin coupling (20 random margins)
CF7	fig. 5: the thimble law against uniform on Q_h; concentration at the leximin
CF8	example 25: one count; the walls' distances; three valuations become two
LG1–LG3	appendix A: the schema; every cited check passes in the log; the rendered table matches `LEDGER.yaml`

References

[1] M. Abreu, Kähler geometry of toric manifolds in symplectic coordinates, in *Symplectic and Contact Topology*, Fields Inst. Commun. **35**, AMS, 2003, 1–24.

[2] S. Amari and H. Nagaoka, *Methods of Information Geometry*, AMS and Oxford University Press, 2000.

[3] V. K. Awasthi, Bridges: from probability and entropy to branes, with worked examples, technical note, Johns Hopkins University, October 2026.

[4] V. K. Awasthi, Coupling polytopes and toric geometry: the 2×3 family, the square case, higher levels and stability, companion paper, Johns Hopkins University, October 2026.

[5] V. K. Awasthi, How the programme fits together: one family of polytopes, many models, and one example through every layer, draft 1, Johns Hopkins University, October 2026.

[6] V. K. Awasthi, Probability, context and differentiation: the layers, working draft, 2026.

[7] M. W. Birch, Maximum likelihood in three-way contingency tables, *J. Roy. Statist. Soc. Ser. B* **25** (1963) 220–233.

[8] Y. M. M. Bishop, S. E. Fienberg and P. W. Holland, *Discrete Multivariate Analysis*, MIT Press, 1975.

[9] J. Bochnak, M. Coste and M.-F. Roy, *Real Algebraic Geometry*, Ergebnisse der Mathematik **36**, Springer, 1998.

[10] C.-H. Cho and Y.-G. Oh, Floer cohomology and disc instantons of Lagrangian torus fibers in Fano toric manifolds, *Asian J. Math.* **10** (2006) 773–814.

[11] T. Delzant, Hamiltoniens périodiques et images convexes de l'application moment, *Bull. Soc. Math. France* **116** (1988) 315–339.

[12] P. Diaconis and B. Sturmfels, Algebraic algorithms for sampling from conditional distributions, *Ann. Statist.* **26** (1998) 363–397.

[13] J. J. Duistermaat and G. J. Heckman, On the variation in the cohomology of the symplectic form of the reduced phase space, *Invent. Math.* **69** (1982) 259–268.

[14] N. Eriksson, S. E. Fienberg, A. Rinaldo and S. Sullivant, Polyhedral conditions for the nonexistence of the MLE for hierarchical log-linear models, *J. Symbolic Comput.* **41** (2006) 222–233.

[15] S. E. Fienberg and A. Rinaldo, Maximum likelihood estimation in log-linear models, *Ann. Statist.* **40** (2012) 996–1023.

[16] R. A. Fisher, The logic of inductive inference, *J. Roy. Statist. Soc.* **98** (1935) 39–82.

[17] K. Fukaya, Y.-G. Oh, H. Ohta and K. Ono, Lagrangian Floer theory on compact toric manifolds I, *Duke Math. J.* **151** (2010) 23–174.

[18] W. Fulton, *Introduction to Toric Varieties*, Annals of Mathematics Studies **131**, Princeton University Press, 1993.

[19] S. Galkin, The conifold point, arXiv:1404.7388, 2014.

[20] S. Galkin, V. Golyshev and H. Iritani, Gamma classes and quantum cohomology of Fano manifolds: Gamma conjectures, *Duke Math. J.* **165** (2016) 2005–2077.

[21] D. Geiger, C. Meek and B. Sturmfels, On the toric algebra of graphical models, *Ann. Statist.* **34** (2006) 1463–1492.

[22] I. M. Gelfand, M. M. Kapranov and A. V. Zelevinsky, *Discriminants, Resultants, and Multidimensional Determinants*, Birkhäuser, 1994.

[23] V. Guillemin, Kaehler structures on toric varieties, *J. Differential Geom.* **40** (1994) 285–309.

[24] K. Hori and C. Vafa, Mirror symmetry, arXiv:hep-th/0002222, 2000.

[25] H. Iritani, An integral structure in quantum cohomology and mirror symmetry for toric orbifolds, *Adv. Math.* **222** (2009) 1016–1079.

[26] J. Judd and K. Rietsch, The tropical critical point and mirror symmetry, arXiv:1911.04463, 2019.

[27] D. Maclagan and B. Sturmfels, *Introduction to Tropical Geometry*, Graduate Studies in Mathematics **161**, AMS, 2015.

[28] R. T. Rockafellar, *Convex Analysis*, Princeton University Press, 1970.

[29] A. Schrijver, *Theory of Linear and Integer Programming*, Wiley, 1986.